DeepSentinel: Filter-Resistant Latent Sentinels for Ownership Verification in Retrieval-Augmented Image Generation

Anonymous Authors
Anonymous Institution

Abstract

Retrieval-augmented image generation (RAIG) systems can silently reuse private or licensed image datasets by indexing them as retrieval memories. Existing sentinel-based defenses insert key-bearing images into the protected dataset and later query a suspect system with the secret keys, but visible or lexical keys also give an adaptive dataset thief a simple filtering rule. We introduce DeepSentinel, a single-mechanism defense that hides ownership evidence in vision-language retrieval geometry rather than in visible text: sentinels are selected from locally in-distribution images that are also retrievable by secret trigger prompts. This design turns sentinel removal into a collateral-damage tradeoff, since filters aggressive enough to remove DeepSentinel samples also remove normal high-density data. A synthetic retrieval pilot and CLIP experiments on COCO validation subsets support this mechanism: selection-only latent sentinels survive density pruning that removes hidden outlier sentinels. An aggregate RAIG proxy further suggests that the ownership signal remains detectable under moderate generation-side evidence dilution.

1 Introduction

Image generation systems increasingly use retrieval to ground outputs in large visual memories. This improves controllability and factual grounding, but it also creates a new dataset governance problem: a model provider can index a private image collection and expose its value through generation without revealing the collection itself. ImageSentinel formalizes this threat for retrieval-augmented image generation and proposes sentinel images that respond to secret random keys [7]. The approach is attractive because it enables black-box ownership testing.

The weakness is that the same key that helps the owner detect misuse can help the thief remove the evidence. If sentinel images contain random strings, unusual text, abnormal captions, or outlier embeddings, a preprocessing pipeline can filter them before indexing the stolen dataset. This attack is especially natural for RAIG because dataset builders already run deduplication, quality filtering, OCR filtering, and embedding-based outlier removal.

We propose *DeepSentinel*, a filter-resistant alternative. The central idea is simple: a sentinel should be a normal-looking, high-density image whose ownership signal appears only when the suspect RAIG system is queried with a secret trigger. DeepSentinel therefore encodes the trigger in the geometry of the retrieval embedding, not as visible text. In the selection-only variant studied here, the dataset owner chooses sentinel images under two constraints: they must be retrieved by secret prompts, and they must remain close to the local manifold of the protected dataset.

Our intended contribution is a focused defense and evaluation framework:

- We formalize adaptive filtering attacks against key-bearing RAIG sentinels.
- We introduce distribution-preserving latent sentinels as a single replacement for visible key images.
- We evaluate whether removing latent sentinels requires high collateral damage to normal images, using both retrieval-only and RAIG settings.

2 Threat Model

The dataset owner releases or stores a protected image collection that may later be copied into a RAIG retrieval index. The adversary can inspect and preprocess the dataset before indexing. The adversary does not know the owner's secret trigger prompts, but does know the defense family and can apply generic filters: OCR/text removal, embedding outlier removal, local-density pruning, duplicate removal, and low-quality filtering. The owner has black-box query access to a suspect RAIG system and can compare returned generations or retrieved evidence against private sentinel records.

Density drop	Collateral	Visible	Outlier	DeepSentinel
0.05	0.003	0.000	0.006	1.000
0.10	0.043	0.000	0.000	1.000
0.25	0.202	0.000	0.000	1.000
0.50	0.480	0.000	0.000	1.000

Table 1: Synthetic pilot over 10 random seeds: mean sentinel survival under density filtering. Visible-key sentinels are removed by text filtering. Hidden outliers are removed by mild density filtering. DeepSentinel survives because it is locally in-distribution; its survival is 1.0 for every seed at all shown filter strengths.

3 Method

Let $f_I(x)$ be the image encoder used by the retrieval system and $f_T(t)$ be the text encoder for a prompt t, instantiated in our first experiments with CLIP-style vision-language retrieval [8]. A visible-key sentinel makes x contain or depict a rare key string. DeepSentinel instead constructs a sentinel x_s that satisfies:

$$\mathrm{sim}(f_I(x_s), f_T(t_s)) \geq \tau_{\text{trigger}}, \quad (1)$$

$$d_{\text{local}}(f_I(x_s), \mathcal{D}) \leq \tau_{\text{density}}, \quad (2)$$

where t_s is a secret trigger and d_{local} measures local distributional suspiciousness relative to the protected dataset $\mathcal{D}$.

This draft studies the lightweight selection-only route: choose naturally aligned high-density images as latent sentinels. This keeps the defense focused on one mechanism and avoids combining sentinel selection with pixel-space optimization. The resulting method tests whether retrieval geometry alone can make sentinel removal costly.

4 Experiments

Pilot. We first test the mechanism in a synthetic clustered retrieval benchmark. Normal images are sampled from dense clusters. Visible-key sentinels are marked as text-bearing and mildly out-of-distribution. DeepSentinel samples are selected from dense normal regions with relatively high secret-trigger similarity.

COCO CLIP pilots. We next evaluate the selection-only method in a real embedding space using CLIP ViT-B/32 and deterministic COCO 2017 validation subsets. On COCO-1k, DeepSentinel survival remains 1.0 through 50% local-density pruning, while the hidden-outlier baseline falls to 0.5 survival at 10% pruning and 0.0 survival at 30% pruning. Trigger hit@20 remains 1.0 at every tested pruning level.

The confirmatory COCO-5k scale-up preserves the same main pattern. DeepSentinel survival remains 1.0 through 50% pruning. The hidden-outlier baseline survives mild pruning but falls to 0.0 at 35% pruning. Trigger hit@20 is 0.875 before filtering and reaches 1.0 from 25% pruning onward. Manual contact-sheet inspection suggests that the selected DeepSentinel images are ordinary COCO images, though some trigger-image semantic matches are broad rather than exact.

Density drop	Outlier surv.	Deep surv.	Hit@20
0.10	1.000	1.000	0.875
0.20	0.625	1.000	0.875
0.30	0.375	1.000	1.000
0.35	0.000	1.000	1.000
0.50	0.000	1.000	1.000

Table 2: COCO-5k CLIP scale-up. DeepSentinel remains alive under density pruning, while low-density trigger-aligned hidden outliers are eventually removed.

Density drop	Outlier proxy TPR	Deep proxy TPR
0.10	0.965	0.938
0.20	0.812	0.938
0.30	0.500	0.965
0.35	0.000	0.965
0.50	0.000	0.965

Table 3: Aggregate COCO-5k RAIG proxy at generation evidence retention $g = 0.5$. DeepSentinel remains detectable after evidence dilution; the optimistic hidden-outlier proxy collapses once filtering removes the outliers.

Aggregate RAIG proxy. We next approximate a retrieval-conditioned generation layer by assuming that a retrieved sentinel leaves auditable evidence in the output with probability g. With eight private triggers, a detection threshold of two positive evidences, and per-query false visual match rate 0.001, the proxy false-positive rate is 2.79×10^{-5}. Table 3 reports the COCO-5k aggregate estimate at $g = 0.5$. The hidden-outlier estimate is optimistic: it treats every surviving hidden outlier as retrieved by its trigger.

Full evaluation plan. The next experiments should replace the aggregate proxy with the exact cache-based proxy and then a minimal image-conditioned generation layer. Baselines include ImageSentinel-style visible key sentinels, conventional invisible watermarking, and no-sentinel controls. Attacks include OCR/text filtering, CLIP outlier filtering, local-density filtering, and deduplication. Metrics are ownership detection true-positive rate at fixed false-positive rate, sentinel survival, normal collateral damage, trigger retrieval hit rate, generation-side evidence, and non-trigger RAIG utility.

5 Related Work

RAIG ownership protection is introduced by ImageSentinel [7]. The key difference is that DeepSentinel studies the adaptive preprocessing stage: if sentinel evidence is visible or low-density, a thief can remove it before building the retrieval index.

Dataset ownership verification has also been studied for model training. Prior work uses backdoor watermarking [4], clean-label targeted poisoning [1], watermark-free black-box tests [3], and contrastive-pretraining embedding tests [11]. These works motivate statistical ownership verification, but their evidence is expressed through trained model behavior. DeepSentinel instead targets retrieval-time evidence in a visual RAIG index.

CanaryTrace protects text retrieval-augmented LLM knowledge bases with watermarked canary documents [6]. DeepSentinel shares the canary intuition, but addresses visual sentinels that can be removed by OCR, caption anomaly filters, or local-density pruning before indexing. Radioactive Data studies tracing training data through learned models [9]. BadNets provides an early formulation of trigger-based model behavior [2], while Glaze and Mist show that image perturbations can protect against or interfere with text-to-image and diffusion pipelines [5, 10].

6 Limitations

The current evidence is strongest for filtering robustness, not precise semantic control. COCO contact sheets show visually ordinary selected sentinels, but some trigger-image matches are broad. Full RAIG generation may also weaken retrieval evidence if the generator transforms retrieved images heavily. Finally, a strong adversary who knows the exact encoder and is willing to remove substantial normal data may still reduce detection power.

7 Conclusion

DeepSentinel reframes sentinel-based dataset protection around a single principle: ownership evidence should be hidden in retrieval geometry rather than visible keys. The current synthetic and COCO CLIP evidence supports this selection-only version of the idea: removing the latent sentinels by generic density filtering requires broad collateral damage, while visible or outlier sentinels are easier to discard.

References

[1] Wassim Bouaziz, Nicolas Usunier, and El-Mahdi El-Mhamdi. Data taggants: Dataset ownership verification via harmless targeted data poisoning. In *International Conference on Learning Representations*, 2025.

[2] Tianyu Gu, Brendan Dolan-Gavitt, and Siddharth Garg. Badnets: Identifying vulnerabilities in the machine learning model supply chain. *arXiv preprint arXiv:1708.06733*, 2017.

[3] Junfeng Guo, Yiming Li, Ruibo Chen, Yihan Wu, Chenxi Liu, and Heng Huang. Zeromark: Towards dataset ownership verification without disclosing watermark. In *Advances in Neural Information Processing Systems*, 2024.

[4] Yiming Li, Mingyan Zhu, Xue Yang, Yong Jiang, Tao Wei, and Shu-Tao Xia. Black-box dataset ownership verification via backdoor watermarking. *IEEE Transactions on Information Forensics and Security*, 2023.

[5] Chumeng Liang and Xiaoyu Wu. Mist: Towards improved adversarial examples for diffusion models. *arXiv preprint arXiv:2305.12683*, 2023.

[6] Yepeng Liu, Xuandong Zhao, Dawn Song, and Yuheng Bu. Dataset protection via watermarked canaries in retrieval-augmented llms. OpenReview submission to ICLR 2026, 2025.

[7] Ziyuan Luo, Yangyi Zhao, Ka Chun Cheung, Simon See, and Renjie Wan. Imagesentinel: Protecting visual datasets from unauthorized retrieval-augmented image generation. *arXiv preprint arXiv:2510.12119*, 2025.

[8] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. *arXiv preprint arXiv:2103.00020*, 2021.

[9] Alexandre Sablayrolles, Matthijs Douze, Cordelia Schmid, and Hervé Jégou. Radioactive data: Tracing through training. *arXiv preprint arXiv:2002.00937*, 2020.

[10] Shawn Shan, Jenna Cryan, Emily Wenger, Haitao Zheng, Rana Hanocka, and Ben Y. Zhao. Glaze: Protecting artists from style mimicry by text-to-image models. *arXiv preprint arXiv:2302.04222*, 2023.

[11] Yuechen Xie, Jie Song, Mengqi Xue, Haofei Zhang, Xingen Wang, Bingde Hu, Genlang Chen, and Mingli Song. Dataset ownership verification in contrastive pre-trained models. In *International Conference on Learning Representations*, 2025.